INNOVATIVE TALENTS
SCIENCE TRAINING
PROGRAM

2024 AUTUMN QUARTER

Here is to certify that TianYu Wang has attended the course: Motion Structures: Deployable Assemblies of Mechanisms from October 11th 2024 to January 5th 2025.

OFFICIAL TRANSCRIPT

Program	Professor	Class Period	Grade
Motion Structures: Deployable Assemblies of Mechanisms	Zhong You	10/11/2024~01/05/2025	93.00

The curriculum design of the course focuses on drawing lessons from the educational concepts of both General Education and Research-Based Learning of world-class universities.

Professor: Zhong You
Home Institution: University of Oxford

GRADE	EQUIVALENT PERCENTAGES	GRADE	EQUIVALENT PERCENTAGES
A+	90-100	C+	67-69
A	85-89	C	63-66
A-	80-84	C-	60-62
B+	77-79	D+	57-59
B	73-76	D	53-56
B-	70-72	D-	50-52

Other Grading Information: Nonacademic Credit=Attended, Audited.

For more information visit https://student.neoschool.com/#/passport/login and go to the Course details-Grade page in the Classroom Section.

Please note: The course syllabus and outlines are strictly in consistent with professor's home institutions. All lectures and readings are in English and all students works are also performed in English. Academy consistency is therefore maintained in accordance with the academic requirements at their respective colleges.

Neoscholar Education Group Inc. General Corporation SEAL 2020

ISSUED TO:

Name: TianYu Wang **Student ID:** 2220233256*

Class Period: 54 class hours

*Transcript valid only if bearing the Professor's Signature.

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Motion Structures: Deployable Assemblies of Mechanisms

2024 .ITSTP

Basic Information

Course Title	Motion Structures: Deployable Assemblies of Mechanisms
Instructor	Professor Zhong You, Ph.D. Professor of Engineering Science, University of Oxford
Prerequisites	Mathematics: Vectors, matrices and transformations, complex numbers. Preferable to have used MATLAB (not compulsory); Physics: Statics (equilibriums of forces and moments), dynamics (velocity and accelerations).
Required Text & Tools	All material will be provided in class in the form of PowerPoints.
Grading Criteria	10% Participation 40% Homework 50% Exam (Mid-term & Final Assignment)
Course Key Words	Materials Science and Engineering , Aerospace Engineering , Mechanical Engineering , Structural Mechanics , Foldable Space Structures , Mechanical Components , Origami Materials

Schedule

No.	Topics
Lecture 1	1. Structure Mechanism: Structure mechanism introduction. Mobility calculation of a mechanism. Name and describe various mechanisms and appreciate their uses.
Lecture 2	2. Motion Structures: Motion structures. Applications in aerospace, robotics, civil engineering, biomedical engineering and mechanical engineering.
Lecture 3	3. Planar motion structures: Constructed from 2D mechanisms.
Lecture 4	4. Applications of planar motion structures: Potential applications as retractable roofs
Lecture 5	5. 3D motion structures constructed from 2D mechanisms: Construct from 3D mechanisms using 2D mechanisms as units.
Lecture 6	6. Applications of 3D motion structures: Introduce potential applications as support frames for space antennas.
Lecture 7	7. 3D deployable structures constructed from 3D mechanisms: Spatial linkages. Geometrical conditions of the Bennett linkage and their combinations.
Lecture 8	8. Applications of 3D deployable structures: Alternative forms. Potential applications.
Lecture 9	9. Origami structures: Mathematical principles in origami structures. Mechanical modelling of rigid origami.
Lecture 10	10. Origami structure design and applications: Deployable solar panels. Reflect arrays. Potential applications in robotics.